

Hypotheses testing for single parameter – some common tests

Chi-square – some properties

Theorem 4.2.3 *Let $X_1, \dots, X_k$ be independent χ^2 – random variables with $n_1, \dots, n_k$ degrees of freedom, respectively. Then the sum $V = \sum_{i=1}^k X_i$ is chi-square distributed with $n_1 + n_2 + \dots + n_k$ degrees of freedom.*

Theorem 4.2.4 *Let X_1 and X_2 be independent random variables. Suppose that X_1 is χ^2 with n_1 degrees of freedom, whereas $Y = X_1 + X_2$ is chi-square with n degrees of freedom, where $n > n_1$. Then $X_2 = Y - X_1$ is a chi-square random variable with $n - n_1$ degrees of freedom.*

Theorem 4.2.5 *If a random variable X has a gamma distribution with parameters α and β , then*

$$Y = \frac{2X}{\beta} \sim \chi^2(2\alpha).$$

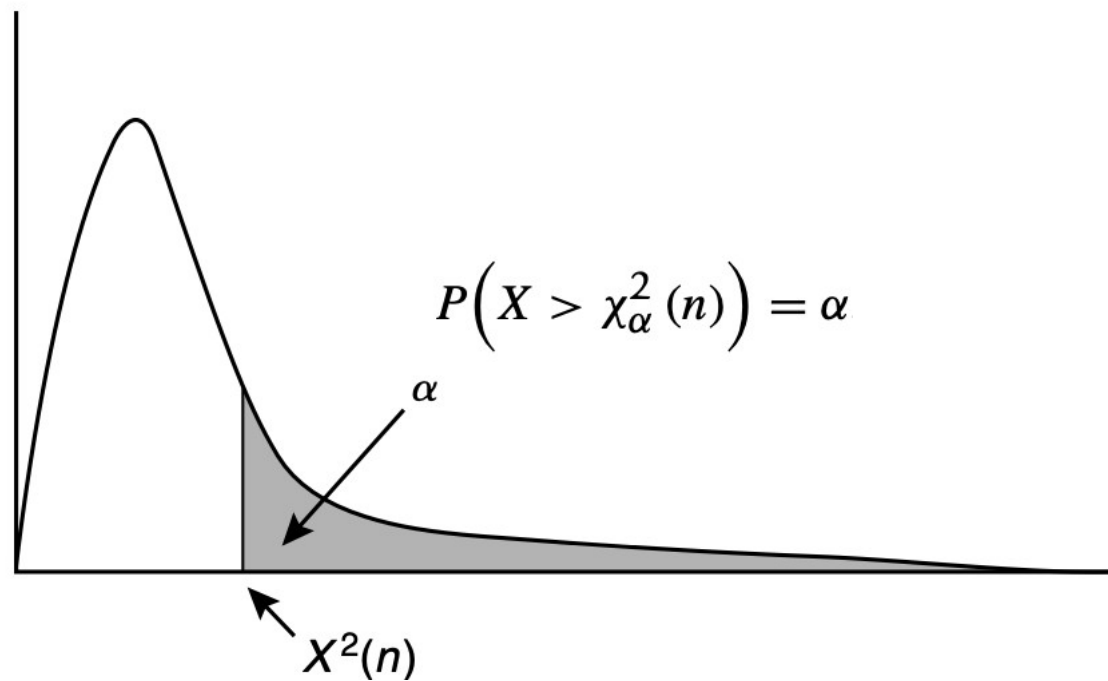
Chi-square – some properties

Theorem 4.2.7 *Let the random sample $X_1, \dots, X_n$ be from a $N(\mu, \sigma^2)$ distributed. Then $Z_i = (X_i - \mu)/\sigma, i = 1, \dots, n$ are independent standard normal random variables and*

$$\sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2$$

has a χ^2 -distribution with n degrees of freedom. In particular, if $X_1, \dots, X_n$ are independent standard normal random variables, then $Y^2 = \sum_{i=1}^n X_i^2$ is chi-square distributed with n degrees of freedom.

Chi-square – some properties



Chi-square – some properties

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Theorem 4.2.8 *If $X_1, \dots, X_n$ is a random sample from a normal population with the mean μ and variance σ^2 , then*

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} = \frac{(n-1) S^2}{\sigma^2}.$$

has a chi-square distribution with $(n-1)$ degrees of freedom.

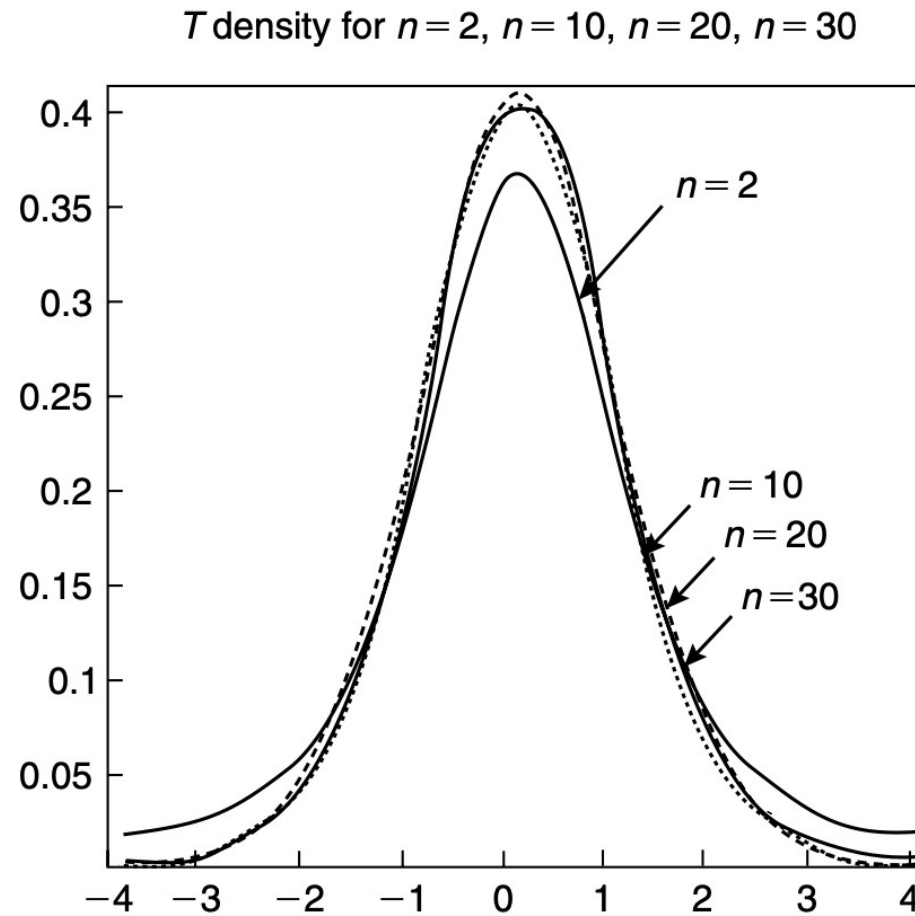
t-distribution – some properties

Definition 4.2.2 *If Y and Z are independent random variables, Y has a chi-square distribution with n degrees of freedom, and $Z \sim N(0, 1)$, then*

$$T = \frac{Z}{\sqrt{Y/n}}$$

*is said to have a (Student) **t-distribution** with n degrees of freedom. We denote this by $T \sim T_n$.*

t-distribution – some properties



For $n \geq 30$ one can approximate t -distribution using standard normal

■ **FIGURE 4.3** The Student t -distribution.

t-distribution – some properties

Theorem 4.2.9 *If $\bar{X}$ and S^2 are the mean and the variance of a random sample of size n from a normal population with the mean μ and variance σ^2 , then*

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t -distribution with $(n - 1)$ degrees of freedom.

CLT

Theorem 4.4.1 *Suppose that the population (not necessarily normal) from which samples are taken has a probability distribution with mean μ and variance σ^2 . Then the standardized variable (or z-transform) associated with $\bar{X}$, given by*

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

is asymptotically standard normal. That is,

$$\lim_{n \rightarrow \infty} P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

CLT –binomial approximation

The fraction of successes in n trials is

$$\frac{Y}{n} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

Hence, Y/n is a sample mean. Since $E(X_i) = p$ and $Var(X_i) = p(1-p)$, we have

$$E\left(\frac{Y}{n}\right) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} np = p$$

$$Var\left(\frac{Y}{n}\right) = \frac{1}{n^2} \sum_{i=1}^n Var(X_i) = \frac{p(1-p)}{n}.$$

$$\frac{Y}{n} \sim \mathcal{N}\left(p, \frac{p(1-p)}{n}\right)$$

*for large sample size usually $np \geq 5$
and $n(1-p) \geq 5$.*

Test about population mean

SUMMARY OF HYPOTHESIS TESTS FOR μ

Large Sample ($n \geq 30$)

To test

$$H_0 : \mu = \mu_0$$

versus

$$\mu > \mu_0, \text{ upper tail test}$$

$$H_a : \mu < \mu_0, \text{ lower tail test}$$

$$\mu \neq \mu_0, \text{ two-tailed test}$$

$$\text{Test statistic: } Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Replace σ by S , if σ is unknown.

$$\text{Rejection region : } \begin{cases} z > z_{\alpha}, & \text{upper tail RR} \\ z < -z_{\alpha}, & \text{lower tail RR} \\ |z| > z_{\alpha/2}, & \text{two tail RR} \end{cases}$$

Small Sample ($n < 30$)

To test

$$H_0 : \mu = \mu_0$$

versus

$$\mu > \mu_0, \text{ upper tail test}$$

$$H_a : \mu < \mu_0, \text{ lower tail test}$$

$$\mu \neq \mu_0, \text{ two-tailed test}$$

$$\text{Test statistic: } T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

$$\text{RR : } \begin{cases} t > t_{\alpha, n-1}, & \text{upper tail RR} \\ t < -t_{\alpha, n-1}, & \text{lower tail RR} \\ |t| > t_{\alpha/2, n-1}, & \text{two tail RR} \end{cases}$$

For small sample case we use normal plots to check for normality

Test about population mean

Assumption: $n \geq 30$

Assumption: Random sample comes from a normal population

Decision: Reject H_0 , if the observed test statistic falls in the RR and conclude that H_a is true with $(1 - \alpha)100\%$ confidence. Otherwise, keep H_0 so that there is not enough evidence to conclude that H_a is true for the given α and more experiments may be needed.

Test about population mean

Example 7.4.6

A machine is considered to be unsatisfactory if it produces more than 8% defectives. It is suspected that the machine is unsatisfactory. A random sample of 120 items produced by the machine contains 14 defectives. Does the sample evidence support the claim that the machine is unsatisfactory? Use $\alpha = 0.01$.

Solution

Let Y be the number of observed defectives. This follows a binomial distribution. However, because np_0 and nq_0 are greater than 5, we can use a normal approximation to the binomial to test the hypothesis. So we need to test $H_0 : p = 0.08$ versus $H_a : p > 0.08$. Let the point estimate of p be $\hat{p} = (Y/n) = 0.117$, the sample proportion. Then the value of the TS is

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.117 - 0.08}{\sqrt{\frac{(0.08)(0.92)}{120}}} = 0.137.$$

For $\alpha = 0.01$, $z_{0.01} = 2.33$. Hence, the rejection region is $\{z > 2.33\}$.

Test about proportion

SUMMARY OF LARGE SAMPLE HYPOTHESIS TEST FOR p

To test

$$H_0 : p = p_0$$

versus

$$p > p_0, \quad \text{upper tail test}$$

$$H_a : p < p_0, \quad \text{lower tail test.}$$

Test statistic:

$$Z = \frac{\hat{p} - p_0}{\sigma_{\hat{p}}}, \quad \text{where } \sigma_{\hat{p}} = \sqrt{\frac{p_0 q_0}{n}}, \quad \text{where } q_0 = 1 - p_0.$$

$$\text{Rejection region : } \begin{cases} z > z_{\alpha}, & \text{upper tail RR} \\ z < -z_{\alpha}, & \text{lower tail RR} \\ |z| > z_{\alpha/2}, & \text{two tail RR,} \end{cases}$$

where z is the observed test statistic.

Assumption: n is large. A good rule of thumb is to use the normal approximation to the binomial distribution only when np_0 and $n(1 - p_0)$ are both greater than 5.

Decision: Reject H_0 , if the observed test statistic falls in the RR and conclude that H_a is true with $(1 - \alpha)100\%$ confidence. Otherwise, do not reject H_0 because there is not enough evidence to conclude that H_a is true for given α and more data are needed.

Test about proportion

Example 7.4.6

A machine is considered to be unsatisfactory if it produces more than 8% defectives. It is suspected that the machine is unsatisfactory. A random sample of 120 items produced by the machine contains 14 defectives. Does the sample evidence support the claim that the machine is unsatisfactory? Use $\alpha = 0.01$.

Solution

Let Y be the number of observed defectives. This follows a binomial distribution. However, because np_0 and nq_0 are greater than 5, we can use a normal approximation to the binomial to test the hypothesis. So we need to test $H_0 : p = 0.08$ versus $H_a : p > 0.08$. Let the point estimate of p be $\hat{p} = (Y/n) = 0.117$, the sample proportion. Then the value of the TS is

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.117 - 0.08}{\sqrt{\frac{(0.08)(0.92)}{120}}} = 0.137.$$

For $\alpha = 0.01$, $z_{0.01} = 2.33$. Hence, the rejection region is $\{z > 2.33\}$.

Decision: Because 0.137 is not greater than 2.33, we do not reject H_0 . We conclude that the evidence does not support the claim that the machine is unsatisfactory.

Test about variance

SUMMARY OF HYPOTHESIS TEST FOR THE VARIANCE σ^2

To test

$$H_0 : \sigma^2 = \sigma_0^2$$

versus

$$\begin{aligned} &\sigma^2 > \sigma_0^2, \quad \text{upper tail test} \\ H_a : &\sigma^2 < \sigma_0^2, \quad \text{lower tail test} \\ &\sigma^2 \neq \sigma_0^2, \quad \text{two-tailed test.} \end{aligned}$$

Test about variance

Test statistic:

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

where S^2 is the sample variance.

Observed value of test statistic:

$$\frac{(n-1)s^2}{\sigma_0^2}$$

$$\text{Rejection region : } \begin{cases} \chi^2 > \chi_{\alpha, n-1}^2, & \text{upper tail RR} \\ \chi^2 < \chi_{1-\alpha, n-1}^2, & \text{lower tail RR} \\ \chi^2 > \chi_{\alpha/2, n-1}^2 \text{ or } \chi^2 < \chi_{1-\alpha/2, n-1}^2, & \text{two tail RR} \end{cases}$$

where $\chi_{\alpha, n-1}^2$ is such that the area under the chi-square distribution with $(n-1)$ degrees of freedom to its right is equal to α .

Assumption: Sample comes from a normal population.

Decision: Reject H_0 , if the observed test statistic falls in the RR and conclude that H_a is true with $(1 - \alpha)100\%$ confidence. Otherwise, do not reject H_0 because there is not enough evidence to conclude that H_a is true for given α and more data are needed.

Test about variance

Example 7.4.7

A physician claims that the variance in cholesterol levels of adult men in a certain laboratory is at least 100. A random sample of 25 adult males from this laboratory produced a sample standard deviation of cholesterol levels as 12. Test the physician's claim at 5% level of significance.

Solution

To test

$$H_0 : \sigma^2 = 100 \text{ versus } H_a : \sigma^2 < 100$$

for $\alpha = 0.05$, and 24 degrees of freedom, the rejection region is

$$RR = \{\chi^2 < \chi_{1-\alpha, n-1}^2\} = \{\chi^2 < 13.484\}.$$

The observed value of the TS is

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{(24)(144)}{100} = 34.56.$$

Because the value of the test statistic does not fall in the rejection region, we cannot reject H_0 at 5% level of significance. Here, we assumed that the 25 cholesterol measurements follow the normal distribution.